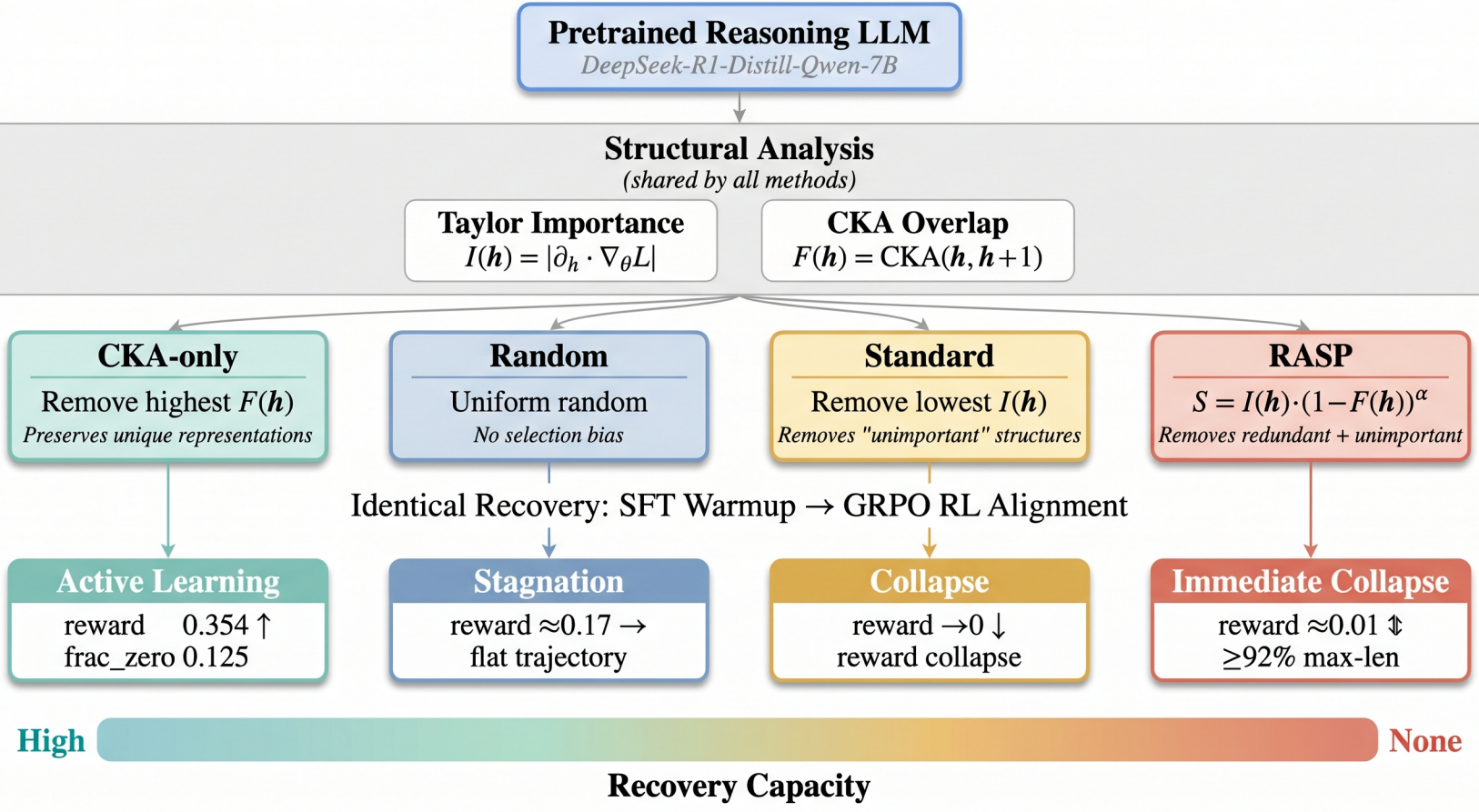

Pretrained Reasoning LLM
DeepSeek-R1-Distill-Qwen-7B
Structural Analysis
(shared by all methods)
Taylor Importance
$I(\boldsymbol{h}) = |\partial_h \cdot \nabla_\theta L|$
CKA Overlap
$F(\boldsymbol{h}) = \mathrm{CKA}(\boldsymbol{h}, \boldsymbol{h}+1)$
CKA-only
Remove highest $F(\boldsymbol{h})$
Preserves unique representations
Random
Uniform random
No selection bias
Standard
Remove lowest $I(\boldsymbol{h})$
Removes "unimportant" structures
RASP
$S = I(\boldsymbol{h}) \cdot (1 - F(\boldsymbol{h}))^{\alpha}$
Removes redundant + unimportant
Identical Recovery: SFT Warmup → GRPO RL Alignment
Active Learning
reward 0.354 ↑
frac_zero 0.125
Stagnation
reward ≈0.17 →
flat trajectory
Collapse
reward →0 ↓
reward collapse
Immediate Collapse
reward ≈0.01 ↕
≥92% max-len
High
None
Recovery Capacity